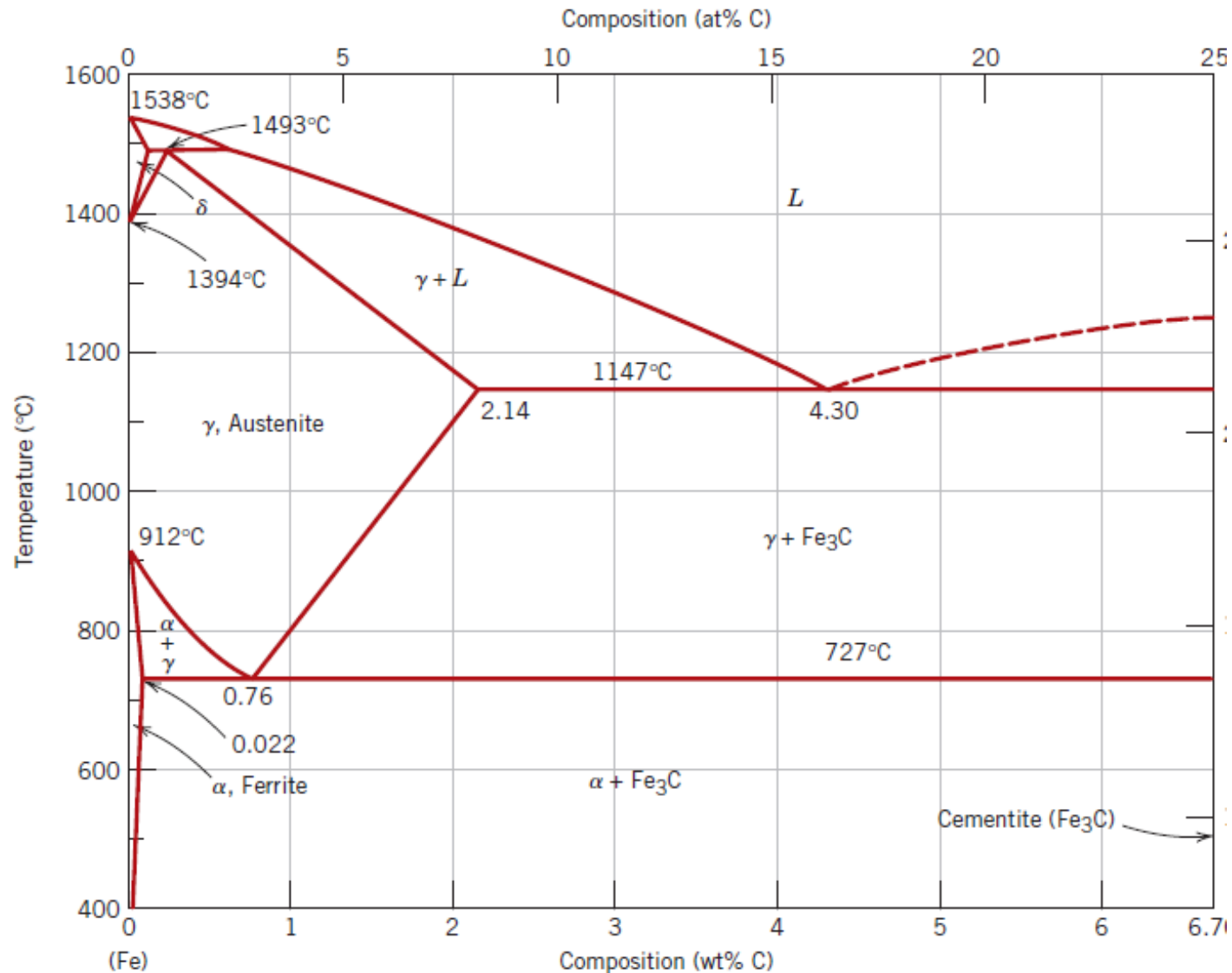
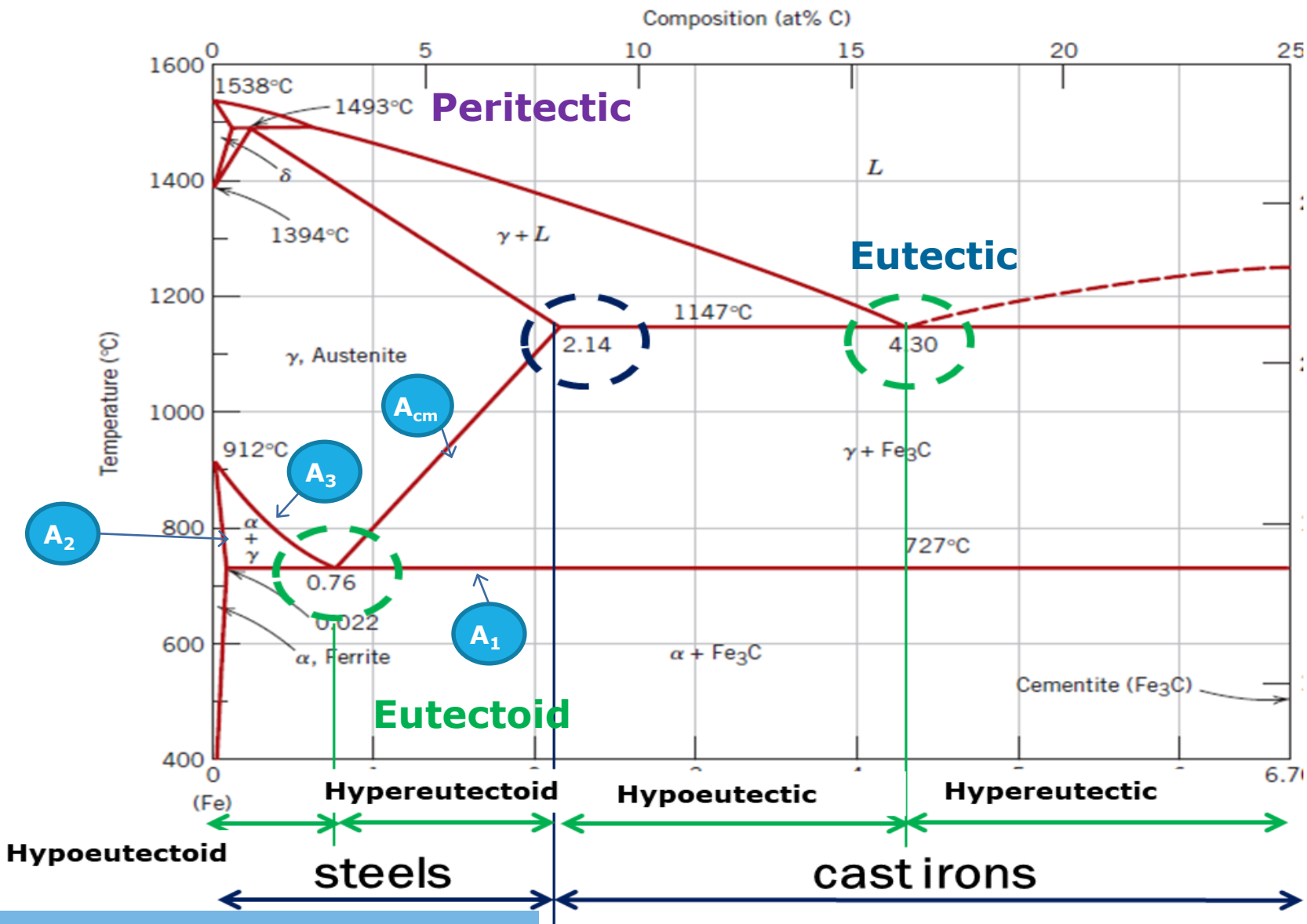


Fe-C phase diagram



Fe-C phase diagram



1.- For a Fe-C alloy with $C_0 = 0.5\%$ of C, at the following temperatures:

$$T_1 = 1147 + \Delta T \text{ } ^\circ\text{C}; T_2 = 1147 - \Delta T \text{ } ^\circ\text{C}; T_3 = 900 \text{ } ^\circ\text{C}; T_4 = 727 + \Delta T \text{ } ^\circ\text{C}; T_5 = 727 - \Delta T \text{ } ^\circ\text{C}$$

a) Indicate the phases present, their chemical composition, and their mass fractions.

b) Indicate the micro constituents, their chemical composition, and their mass fractions.

(Solutions: a) T_1, T_2 , and T_3 : $C_\gamma = 0.5\% \text{ C}$, $w_\gamma = 1$; T_4 : $C_\alpha = 0.022\% \text{ C}$, $C_\gamma = 0.76\% \text{ C}$, $w_\alpha = 0.35$; T_5 : $C_\alpha = 0.022\% \text{ C}$, $C_{Fe3C} = 6.7\% \text{ C}$, $w_\alpha = 0.93$; b) $w_{\alpha'} = 0.35$, $w_{\alpha,E} = 0.58$, $w_{Fe3C} = 0.07$)

2.- For a steel of $C_0 = 0.76\% \text{ C}$, at the following temperatures:

$$T_1 = 1147 + \Delta T \text{ } ^\circ\text{C}; T_2 = 1147 - \Delta T \text{ } ^\circ\text{C}; T_3 = 900 \text{ } ^\circ\text{C}; T_4 = 727 + \Delta T \text{ } ^\circ\text{C}; T_5 = 727 - \Delta T \text{ } ^\circ\text{C}$$

a) Indicate the phases present, their chemical composition, and their mass fractions.

b) Indicate the micro constituents, their chemical composition, and their mass fractions.

(Solutions: a) T_1, T_2, T_3 and T_4 : $C_\gamma = 0.76\% \text{ C}$, $w_\gamma = 1$; T_5 : $C_\alpha = 0.022\% \text{ C}$, $C_{Fe3C} = 6.7\% \text{ C}$, $w_\alpha = 0.89$, $w_{Fe3C'} = 0.11$; b) pearlite $w_{\alpha,E} = 0.89$, $w_{Fe3C,E} = 0.11$)

3.- For an alloy Fe-1.5 wt % C, at the following temperatures :

$$T_1 = 1147 + \Delta T \text{ } ^\circ\text{C}; T_2 = 1147 - \Delta T \text{ } ^\circ\text{C}; T_3 = 900 \text{ } ^\circ\text{C}; T_4 = 727 + \Delta T \text{ } ^\circ\text{C}; T_5 = 727 - \Delta T \text{ } ^\circ\text{C}$$

a) Indicate the phases present, their chemical composition, and their mass fractions.

b) Indicate the micro constituents, their chemical composition, and their mass fractions.

Solutions: a) T_1 and T_2 : $C_\gamma = 1.5\% \text{ C}$, $w_\gamma = 1$; T_3 : $C_{Fe3C} = 6.7\% \text{ C}$, $C_\gamma = 1.3\% \text{ C}$, $w_\gamma = 0.96$, $w_{Fe3C} = 0.04$; T_4 : $C_\gamma = 0.76\% \text{ C}$, $C_{Fe3C} = 6.7\% \text{ C}$, $w_\gamma = 0.87$, $w_{Fe3C} = 0.13$; T_5 : $C_\alpha = 0.022\% \text{ C}$, $C_{Fe3C} = 6.7\% \text{ C}$, $w_\alpha = 0.78$, $w_{Fe3C} = 0.22$; b) Pearlite $w_{Fe3C'} = 0.13$, $w_{pearlite} = 0.87$ $w_{\alpha E} = 0.78$, $w_{Fe3C,E} = 0.09$)

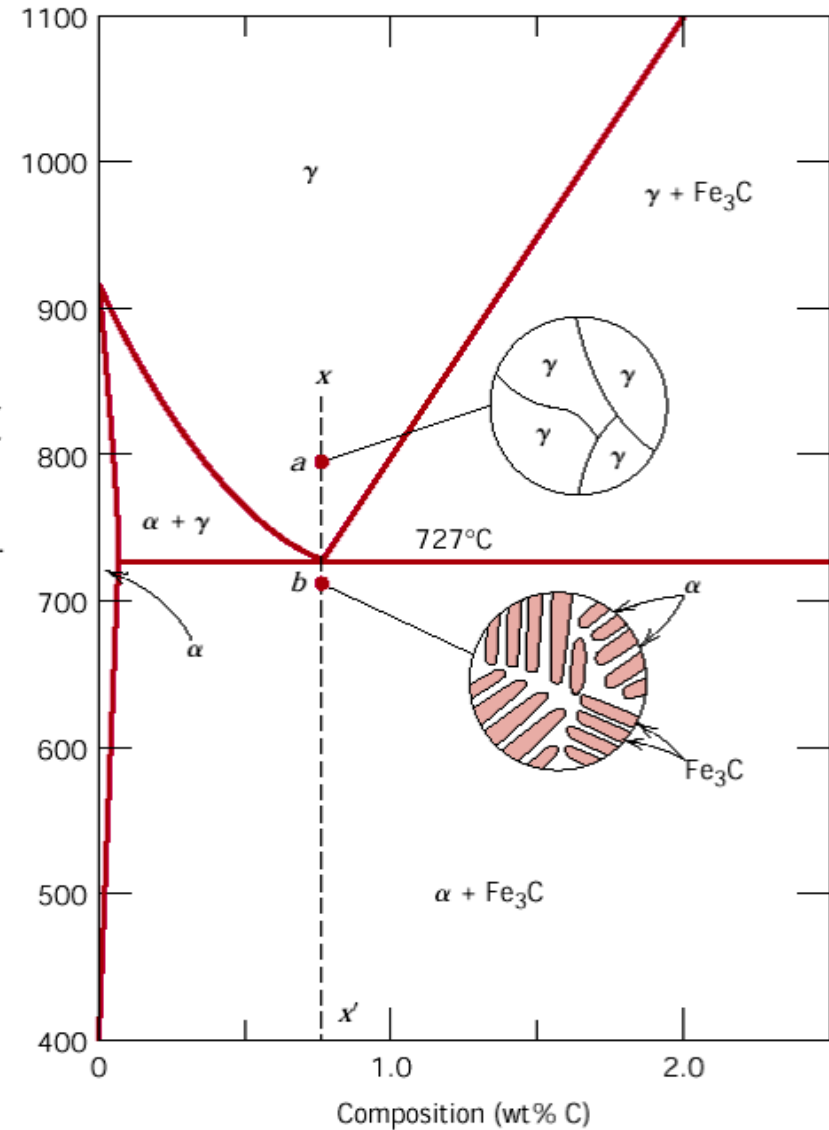
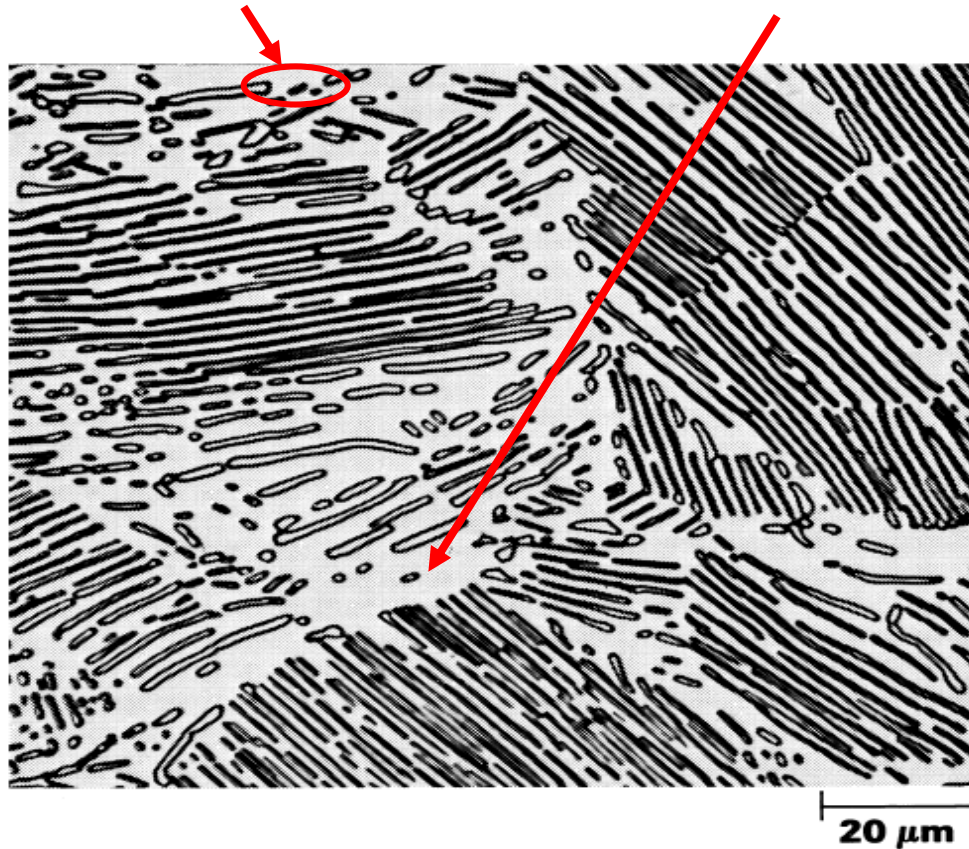
Eutectoid Steels

$$C_0 = 0.76\%C$$

A single microconstituent: **Pearlite**, formed alternating layers of 2 phases:

Cementite (Fe_3C)

Ferrite (αFe)



Eutectoid Steels

$$C_0 = 0.76\%C$$

$T > 727^\circ\text{C}$ (point a)

Phases Present & Composition

Austenite

$$C_{\gamma\text{Fe}} = C_0 = 0.76\%C$$

$T < 727^\circ\text{C}$ (point b)

Phases Present & Composition

α Ferrite & Cementite Fe_3C

$$C_{\text{Fe}-\alpha} = 0.022\%C$$

$$C_{\text{Fe}_3\text{C}} = 6.67\%C$$

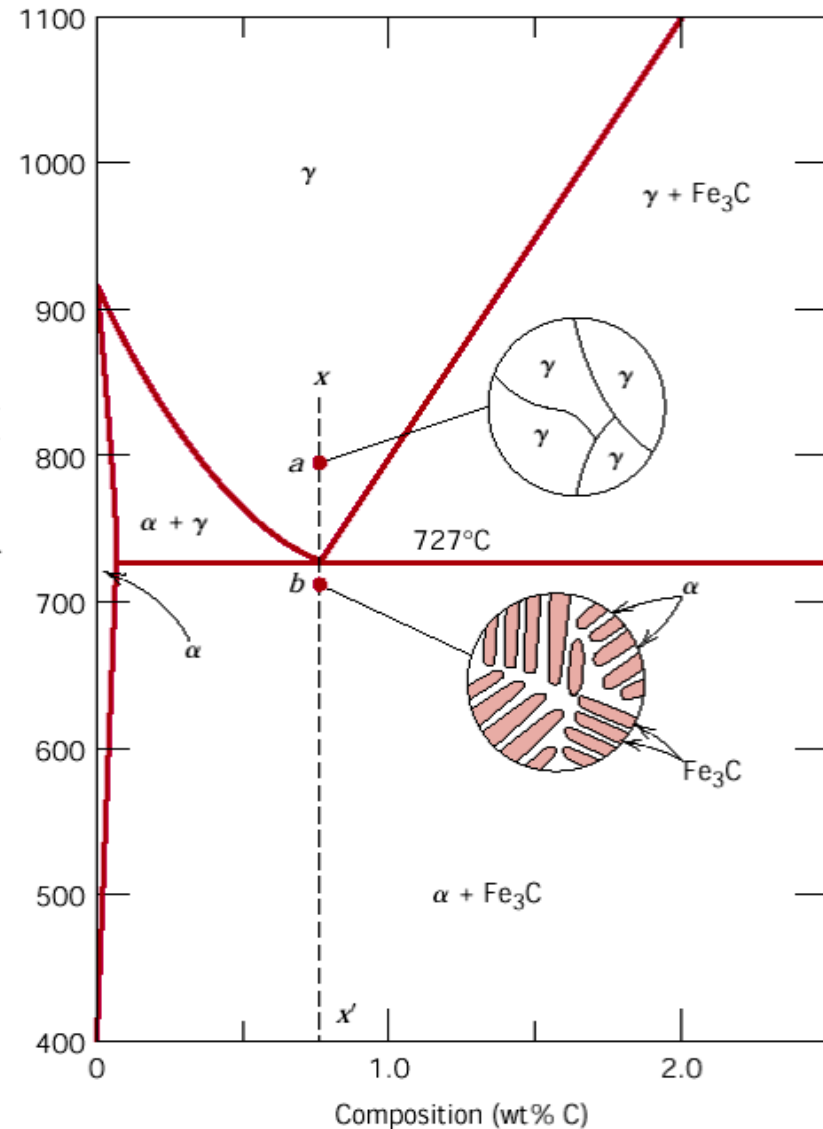
Amount of phases (Weight fractions)

$$\%_{\text{Fe}-\alpha} = \frac{C_{\text{Fe}_3\text{C}} - C_0}{C_{\text{Fe}_3\text{C}} - C_{\alpha}} * 100 = \frac{6.67 - 0.76}{6.67 - 0.022} * 100 = 89\%$$

$$\%_{\text{Fe}_3\text{C}} = 100 - 89 = 11\%$$

Microconstituents

Pearlite



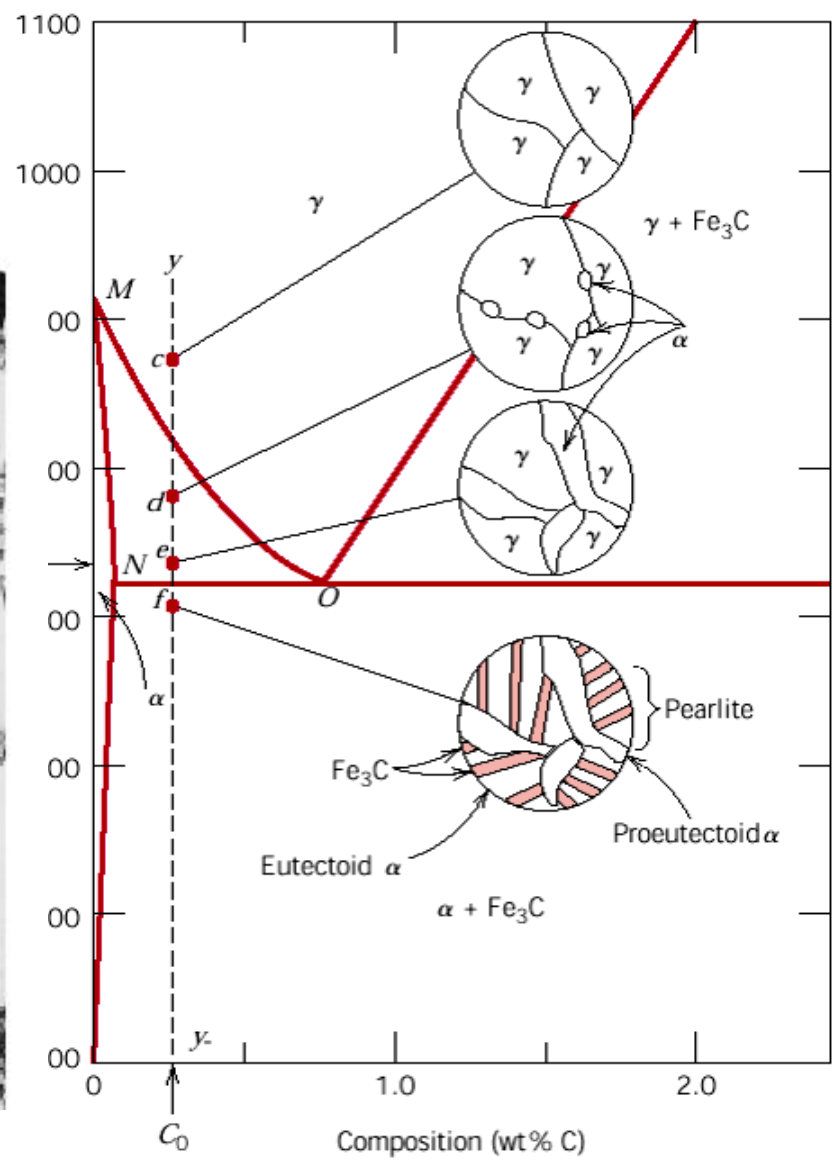
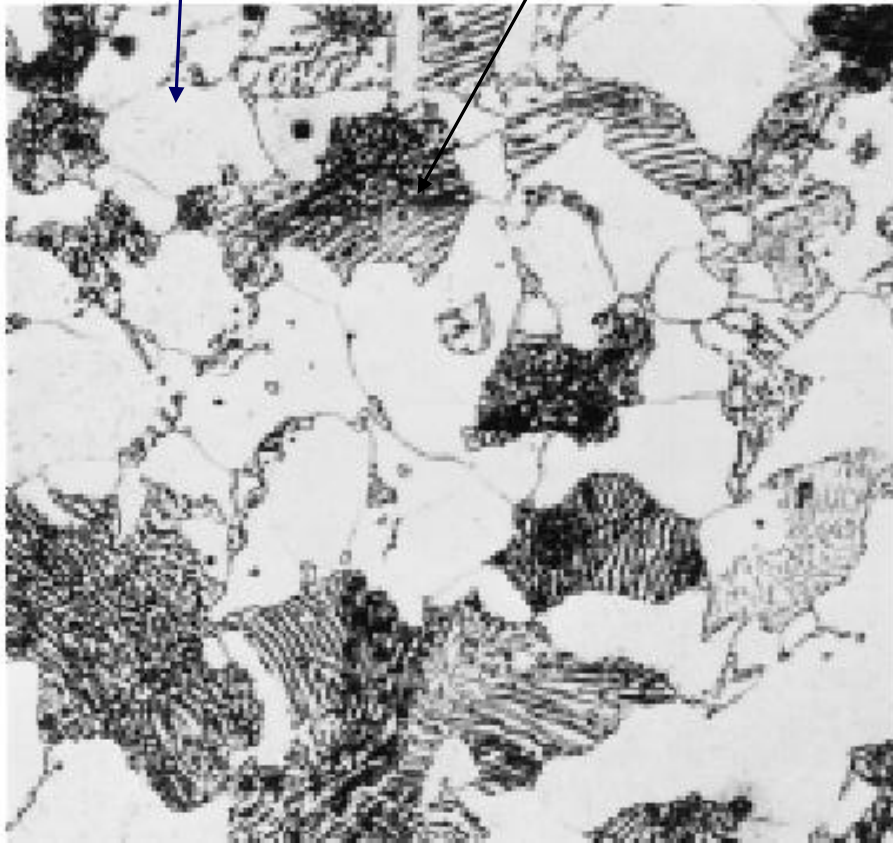
Hypoeutectoid Steels

$C < 0,76\%C$

Two microconstituents

Proeutectoid Ferrite or primary (α Fe)

Pearlite (α Fe + Fe_3C)



Hypoeutectoid Steels

Point c

$$C_0 = 0.3\%C$$

Phases Present & Composition

Austenite (γ)

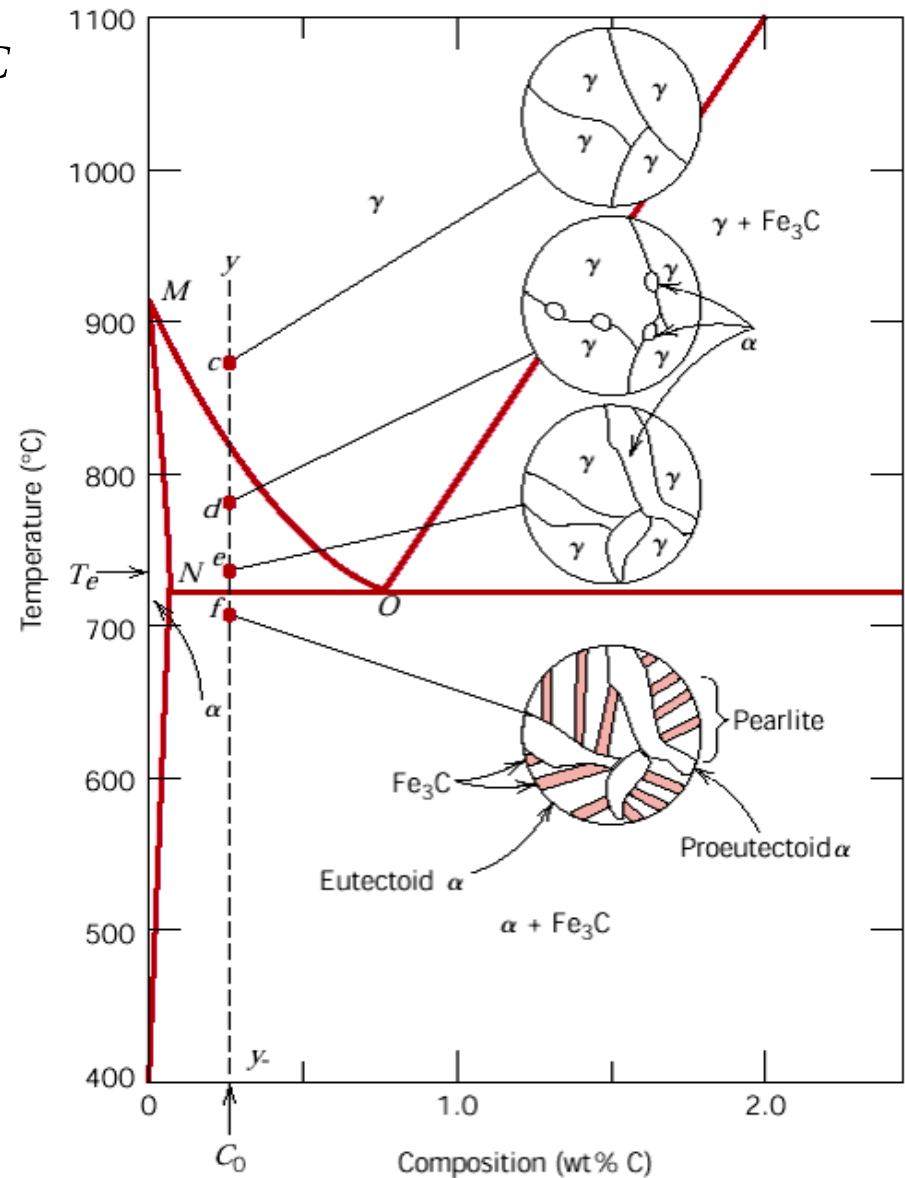
$$C_\gamma = C_0 = 0.3\%C$$

Amount of phases (Weight fractions)

$$\%_{(\gamma)} = 100\%$$

Microconstituents

1 Austenite (γ)



Hypoeutectoid Steels

Point d

$$C_0 = 0.3\%C$$

Phases Present & Composition

Ferrite (α) + Austenite (γ)

$$C_\alpha = 0.02\%C$$

$$C_\gamma = 0.4\%C$$

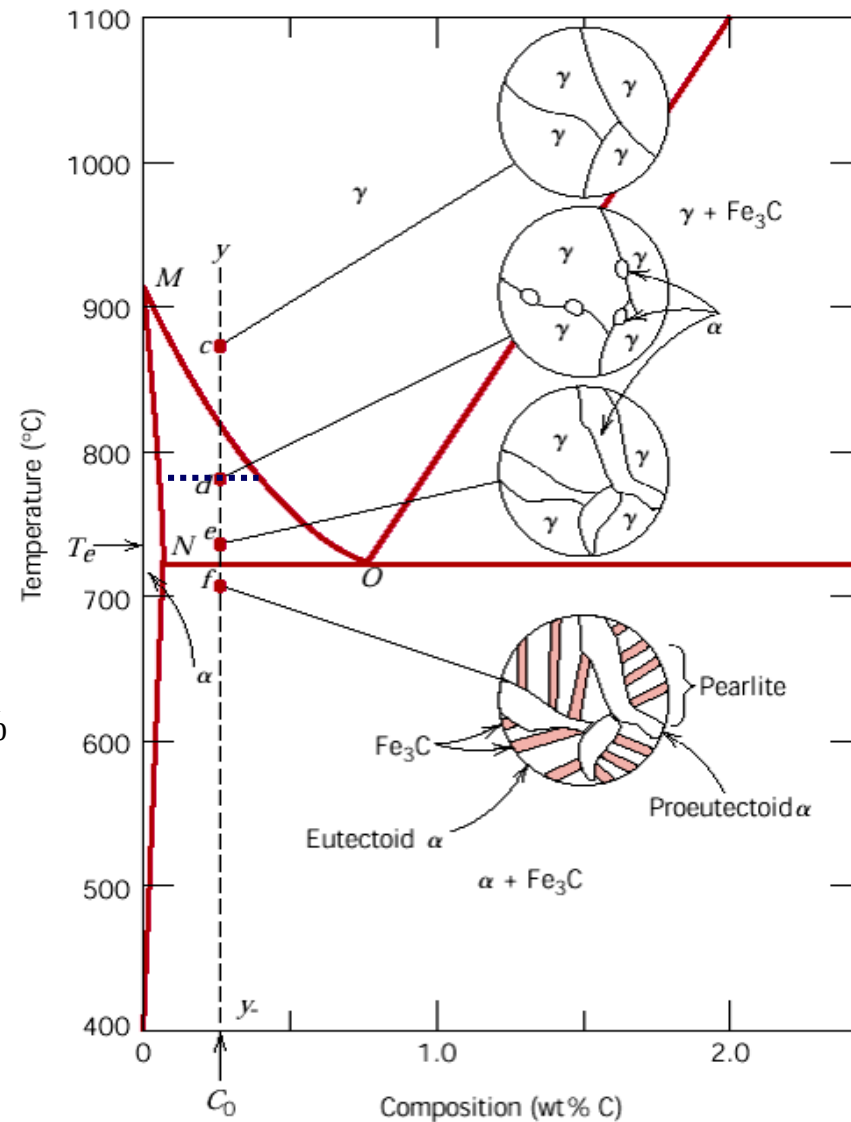
Amount of phases (Weight fractions)

$$\%_\alpha = \frac{S}{R+S} 100 = \frac{C_\gamma - C_0}{C_\gamma - C_{Fe\alpha}} 100 = \frac{0.4 - 0.3}{0.4 - 0.02} 100 = 26\%$$

$$\%_\gamma = \frac{R}{R+S} 100 = \frac{C_0 - C_{Fe\alpha}}{C_\gamma - C_{Fe\alpha}} 100 = \frac{0.3 - 0.02}{0.4 - 0.02} 100 = 74\%$$

Microconstituents

Proeutectoid Ferrite
Austenite



Hypoeutectoid Steels

Point e

$$C_0 = 0.3\%C$$

Phases Present & Composition

Ferrite (α) + Austenite (γ)

$$C_\alpha = 0.02\%C \quad C_\gamma = 0.76\%C$$

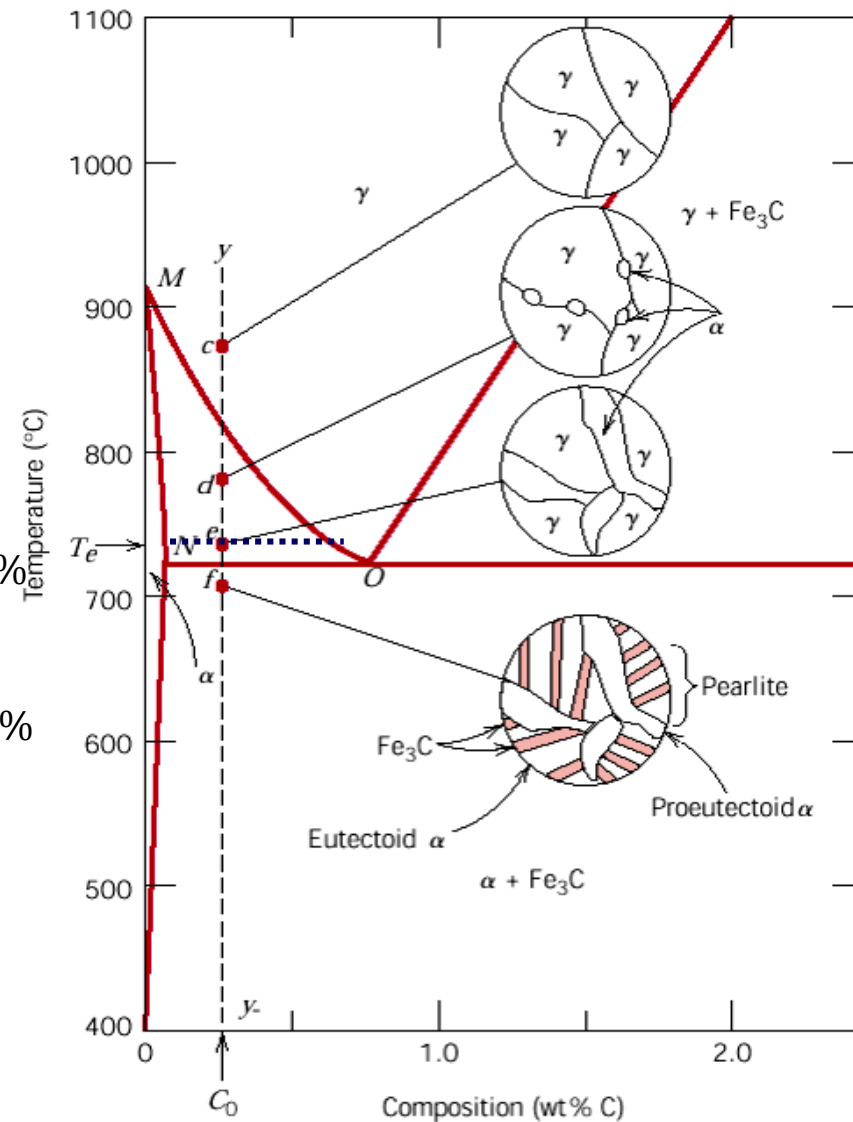
Amount of phases (Weight fractions)

$$\%_{Fe\alpha} = \frac{S}{R+S} 100 = \frac{C_\gamma - C_0}{C_\gamma - C_{Fe\alpha}} 100 = \frac{0.76 - 0.3}{0.76 - 0.02} 100 = 62\%$$

$$\%_{(\gamma)} = \frac{R}{R+S} 100 = \frac{C_0 - C_{Fe\alpha}}{C_\gamma - C_{Fe\alpha}} 100 = \frac{0.3 - 0.02}{0.76 - 0.02} 100 = 38\%$$

Microconstituents

Proeutectoid Ferrite
Austenite



Hypoeutectoid Steels

Point f

$$C_0 = 0.3\%C$$

Phases Present & Composition

Ferrite (α) + Cementite (Fe_3C)

$$C_\alpha = 0.02\%C \quad C_{Fe_3C} = 6.67\%C$$

Amount of phases (Weight fractions)

$$\%_{Fe-\alpha} = \frac{C_{Fe_3C} - C_0}{C_{Fe_3C} - C_\alpha} * 100 = \frac{6.67 - 0.3}{6.67 - 0.022} * 100 = 96\%$$

$$\%_{Fe_3C} = 100 - 96 = 4\%$$

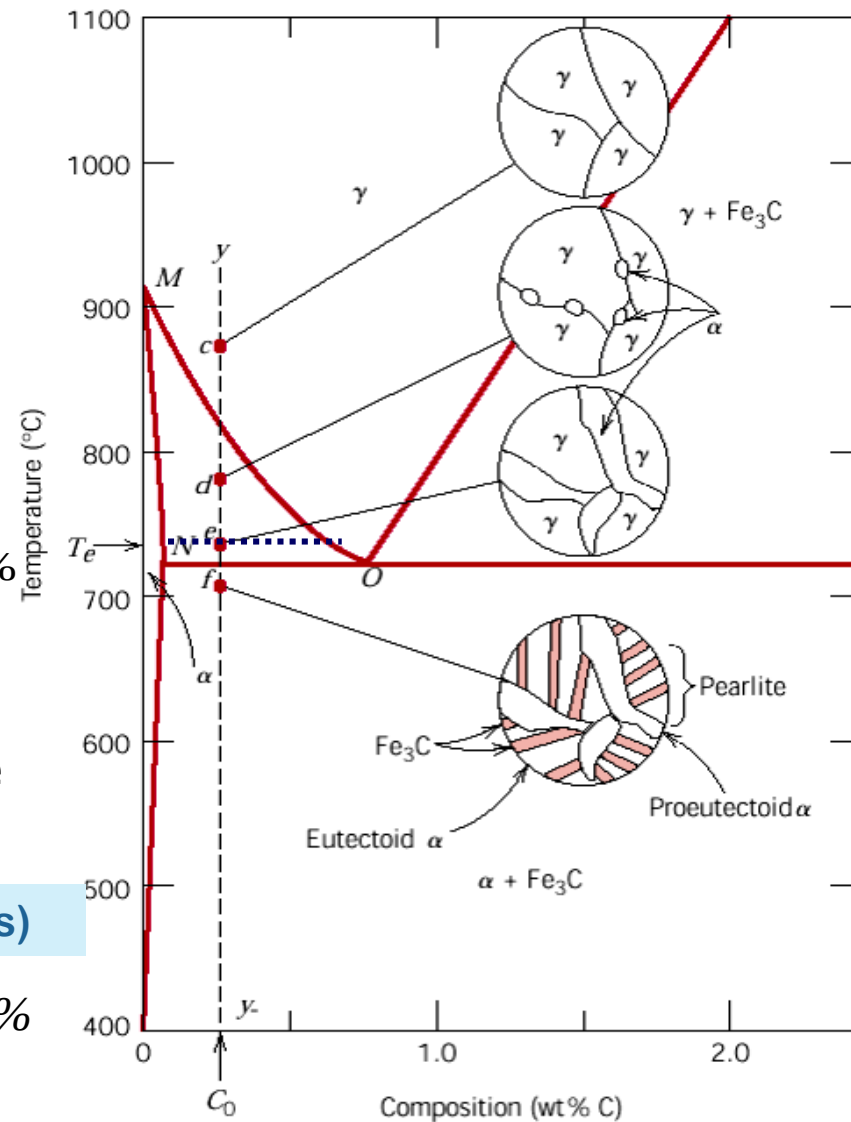
Microconstituents

Proeutectoid Ferrite
Pearlite

Amount of microconstituents (weight fractions)

$$\%_{Fe-\alpha}(P) = \frac{C_{Fe_3C} - C_0}{C_{Fe_3C} - C_\alpha} * 100 = \frac{0.76 - 0.3}{0.76 - 0.022} * 100 = 62\%$$

$$\%_{Fe-\alpha}(E) = 96 - 62 = 34\%$$

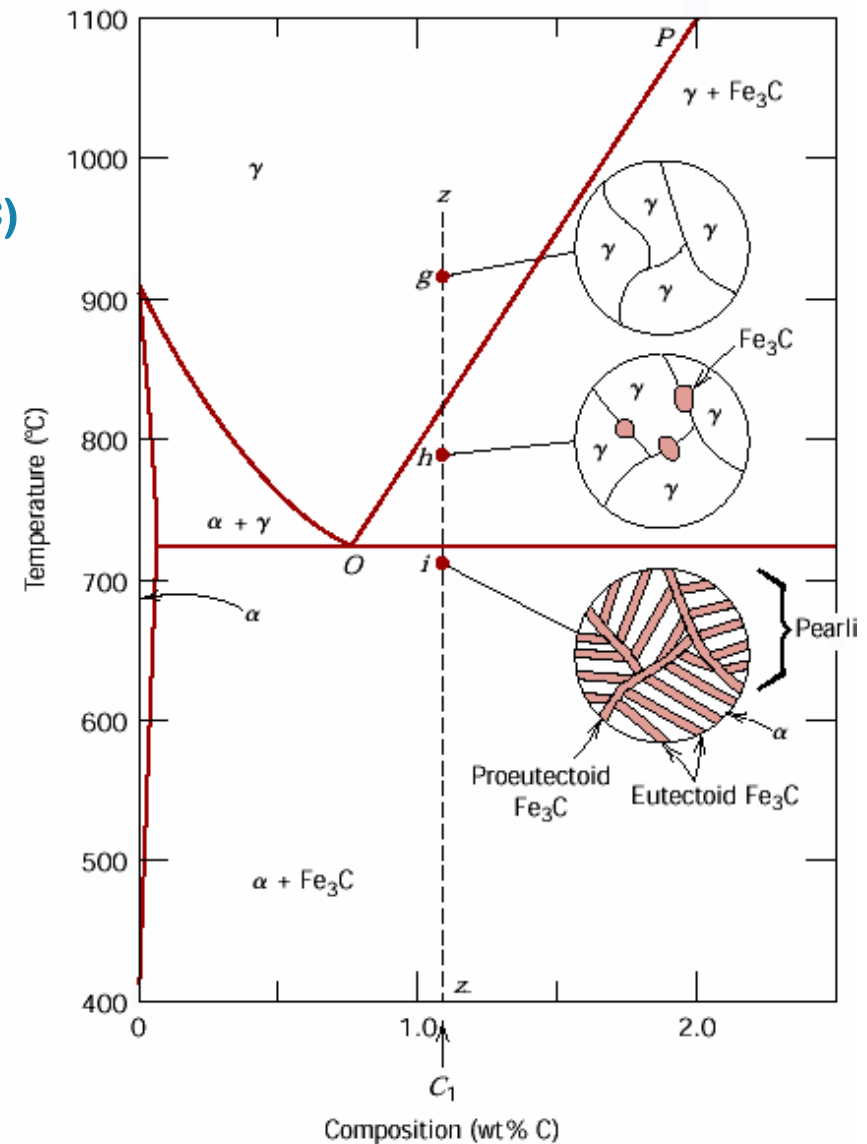
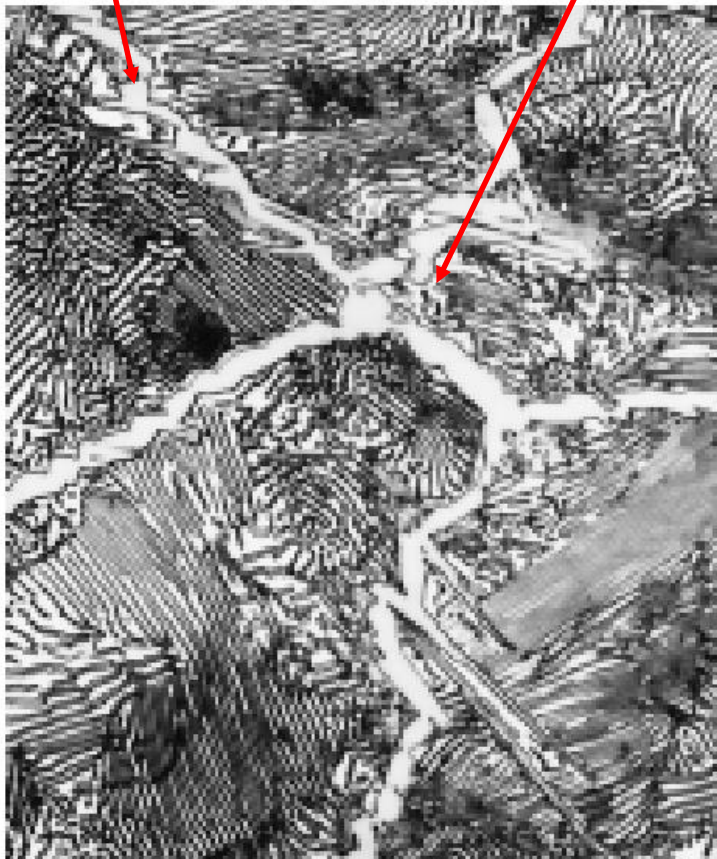


Hypereutectoid Steels

2 microconstituents

Proeutectoid
Cementite Fe_3C

Pearlite ($\text{Fe}_\alpha + \text{Fe}_3\text{C}$)



6. A steel with the content of C < 4,3 % at $727 - \Delta T$ °C. Contains $f_{\alpha} = 0,47$. Indicate the mass fraction of the phases at the temperatures of $1147 + \Delta T$ °C and $1147 - \Delta T$ °C.

$$\chi_{\alpha} = \frac{C_{Fe3C} - C_0}{C_{Fe3C} - C_{\alpha}} = \frac{6.7 - C_0}{6.7 - 0.022} = 0.47 \quad C_0 = 3.6\%$$

$$T1: \chi_L = \frac{C_{Fe3C} - C_0}{C_{Fe3C} - C_{\alpha}} = \frac{3.6 - 2.14}{4.3 - 2.14} = 0.68; \chi_{\gamma} = 1 - 0.68 = 0.32$$

$$T2: \chi_{Fe3C} = \frac{3.6 - 2.14}{6.7 - 2.14} = 0.32; \chi_{\gamma} = 1 - 0.32 = 0.68$$

$$\chi_p = 0.32$$

$$\chi_{\gamma E} = 0.68 - 0.32 = 0.36$$

$$\chi_{Fe3C} = 0.32$$